A Boundary-Sector Counterexample for Conical Frequency Projection

Discovery packet for arXiv:0909.3712

13 August 2026

Status. Full counterexample, pending human review. The boundary extension asked for in Remark 5.2 is false: for every $u, v > 0$ and every $N > 1$ there is $g \in L^2(\mathbb{R}^2)$ such that $(0, v)$ is N-regular for $P_{\mathcal{C}_{u,v}} g$ but not for g.

1 Source question

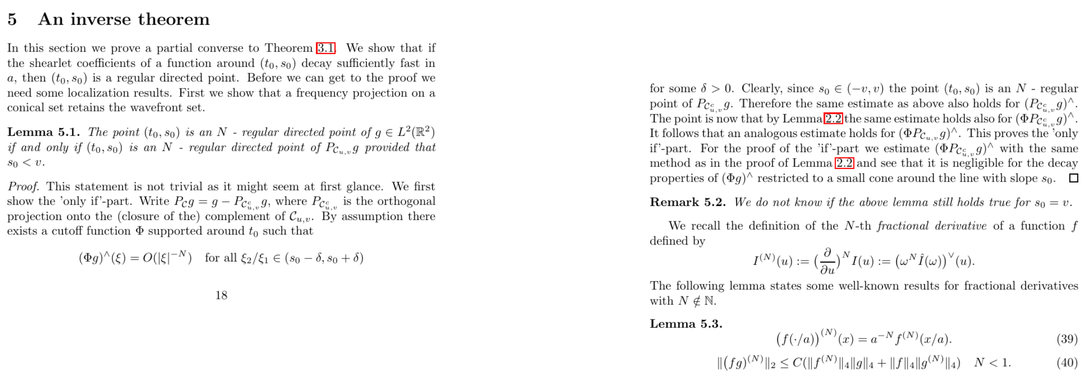
Grohs [1] uses the frequency cone

$$\mathcal{C}_{u,v} = \{\xi \in \mathbb{R}^2 : |\xi_1| \geq u,\ |\xi_2| \leq v|\xi_1|\}$$

and its orthogonal Fourier projection $P_{\mathcal{C}_{u,v}}$. A directed point (t_0, s_0) is N-regular for f when some smooth compactly supported cutoff Φ, equal to one near t_0, and some neighborhood V of s_0 satisfy

$$|\widehat{\Phi f}(\eta)| = O((1 + |\eta|)^{-N}) \quad \text{whenever } \eta_2/\eta_1 \in V.$$

Lemma 5.1 proves that projection preserves and reflects this property at interior slopes $s_0 < v$. Remark 5.2 asks whether the equivalence remains true when $s_0 = v$.



5 An inverse theorem

In this section we prove a partial converse to Theorem 3.1. We show that if the shearlet coefficients of a function around (t_0, s_0) decay sufficiently fast in a, then (t_0, s_0) is a regular directed point. Before we can get to the proof we need some localization results. First we show that a frequency projection on a conical set retains the wavefront set.

Lemma 5.1. *The point (t_0, s_0) is an N - regular directed point of $g \in L^2(\mathbb{R}^2)$ if and only if (t_0, s_0) is an N - regular directed point of $P_{\mathcal{C}_{u,v}} g$ provided that $s_0 < v$.*

Proof. This statement is not trivial as it might seem at first glance. We first show the 'only if'-part. Write $P_{\mathcal{C}} g = g - P_{\mathcal{C}^c_{u,v}} g$, where $P_{\mathcal{C}^c_{u,v}}$ is the orthogonal projection onto the (closure of the) complement of $\mathcal{C}_{u,v}$. By assumption there exists a cutoff function Φ supported around t_0 such that

$$(\Phi g)^\wedge(\xi) = O(|\xi|^{-N}) \quad \text{for all } \xi_2/\xi_1 \in (s_0 - \delta, s_0 + \delta)$$

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for some $\delta > 0$. Clearly, since $s_0 \in (-v, v)$ the point (t_0, s_0) is an N - regular point of $P_{\mathcal{C}^c_{u,v}} g$. Therefore the same estimate as above also holds for $(P_{\mathcal{C}^c_{u,v}} g)^\wedge$. The point is now that by Lemma 2.2 the same estimate holds also for $(\Phi P_{\mathcal{C}^c_{u,v}} g)^\wedge$. It follows that an analogous estimate holds for $(\Phi P_{\mathcal{C}_{u,v}} g)^\wedge$. This proves the 'only if'-part. For the proof of the 'if'-part we estimate $(\Phi P_{\mathcal{C}^c_{u,v}} g)^\wedge$ with the same method as in the proof of Lemma 2.2 and see that it is negligible for the decay properties of $(\Phi g)^\wedge$ restricted to a small cone around the line with slope s_0. □

Remark 5.2. *We do not know if the above lemma still holds true for $s_0 = v$.*

We recall the definition of the N-th *fractional derivative* of a function f defined by

$$I^{(N)}(u) := \left(\frac{\partial}{\partial u}\right)^N I(u) := \left(\omega^N \hat{I}(\omega)\right)^\vee(u).$$

The following lemma states some well-known results for fractional derivatives with $N \notin \mathbb{N}$.

Lemma 5.3.

$$\left(f(\cdot/a)\right)^{(N)}(x) = a^{-N} f^{(N)}(x/a). \tag{39}$$

$$\|(fg)^{(N)}\|_2 \leq C(\|f^{(N)}\|_4\|g\|_4 + \|f\|_4\|g^{(N)}\|_4) \quad N < 1. \tag{40}$$

Figure 1: Lemma 5.1 and the boundary question in the source paper (pp. 18–19).

2 Counterexample

The obstruction is that every slope neighborhood of v meets the exterior of the cone. A slowly decaying Fourier sector there is completely deleted by $P_{\mathcal{C}_{u,v}}$, but its decay survives every spatial localization.

Theorem 1. *Let $u, v > 0$ and $N > 1$. There exists $g \in L^2(\mathbb{R}^2)$ such that $P_{\mathcal{C}_{u,v}} g = 0$, while $(0, v)$ is not an N-regular directed point of g. Consequently the proposed boundary version of Lemma 5.1 is false.*

Proof. Choose α with $1 < \alpha < N$. Let $\rho \in C^\infty([0,\infty))$ satisfy $\rho = 0$ on $[0,1]$ and $\rho = 1$ on $[2,\infty)$. Choose a nonnegative smooth angular function κ on the unit circle which is supported in

$$\{\omega \in S^1 : \omega_1 > 0, \quad v < \omega_2/\omega_1 < v + 1\}$$

(with support understood in the closure) and is strictly positive at every direction whose slope lies in the displayed open interval. Such a function is obtained from the standard flat bump $\exp(-1/((s-v)(v+1-s)))$ in the slope variable.

Define $h(0) = 0$ and, for $\xi \neq 0$,

$$h(\xi) = \rho(|\xi|)|\xi|^{-\alpha}\kappa\left(\frac{\xi}{|\xi|}\right), \qquad g = \mathcal{F}^{-1}h. \tag{1}$$

In polar coordinates,

$$\int_{|\xi|\geq 2} |h(\xi)|^2\, d\xi \leq \|\kappa\|_\infty^2 |S^1| \int_2^\infty r^{1-2\alpha}\, dr < \infty,$$

so $h, g \in L^2(\mathbb{R}^2)$ by Plancherel. The support of h lies outside $\mathcal{C}_{u,v}$ except possibly on a null boundary, where κ vanishes. Therefore

$$\widehat{P_{\mathcal{C}_{u,v}} g} = \mathbf{1}_{\mathcal{C}_{u,v}} h = 0, \qquad P_{\mathcal{C}_{u,v}} g = 0.$$

The projected function is thus N-regular at every directed point.

It remains to prove that g is not N-regular at $(0, v)$. Let Φ be *any* smooth compactly supported cutoff equal to one near zero, and let V be *any* neighborhood of v. Choose $s \in V \cap (v, v+1)$ and set

$$e = \frac{(1, s)}{\sqrt{1 + s^2}}, \qquad c = \kappa(e) > 0.$$

The product–convolution identity gives

$$\widehat{\Phi g}(Re) = \int_{\mathbb{R}^2} \widehat{\Phi}(\eta) h(Re - \eta)\, d\eta. \tag{2}$$

We claim that

$$\lim_{R\to\infty} R^\alpha \widehat{\Phi g}(Re) = c \int_{\mathbb{R}^2} \widehat{\Phi}(\eta)\, d\eta = c\Phi(0) = c. \tag{3}$$

Indeed, on $|\eta| \leq R/2$ one has $|Re - \eta| \geq R/2$; hence $R^\alpha h(Re - \eta)$ is uniformly bounded and converges pointwise to c. Dominated convergence applies against the Schwartz function $\widehat{\Phi}$. On the complementary region, h is bounded and, for every $K > 0$, Schwartz decay gives

$$R^\alpha \int_{|\eta|>R/2} |\widehat{\Phi}(\eta) h(Re - \eta)|\, d\eta \leq C_K R^\alpha \int_{|\eta|>R/2} (1 + |\eta|)^{-K}\, d\eta = O(R^{\alpha+2-K}) = o(1)$$

after choosing $K > \alpha + 2$. This proves (3).

Thus $|\widehat{\Phi g}(Re)| \sim cR^{-\alpha}$. Since $\alpha < N$, this is not $O(R^{-N})$. The ray Re has slope $s \in V$, and both Φ and V were arbitrary. Hence no localization and no slope neighborhood can certify N-regularity of g at $(0, v)$. $\square$

3 Scope and checks

The counterexample destroys the "if" direction at the boundary: $P_{\mathcal{C}_{u,v}} g$ is regular but g is not. It does not affect Lemma 5.1 for strictly interior slopes, where one can choose a slope neighborhood separated from the exterior sector. The construction is genuinely L^2 and uses a smooth angular cutoff flat at the cone boundary; it does not rely on a delta mass or another distribution supported on a measure-zero ray.

The quantifiers in directed regularity are essential. Inspecting h alone would not suffice. Equation (3) proves the obstruction for every admissible spatial cutoff and every neighborhood of the boundary slope. The accompanying script checks the support geometry and integrability exponents; the convolution limit is the analytic heart of the proof.

The paper's other structured-dual question is historically resolved by the author's follow-up [2], which gives a nonexistence obstruction. A bounded local-corpus and web search on 13 August 2026 found no explicit resolution of Remark 5.2 or this sector construction. This is not an exhaustive novelty claim. Human review should focus on the localization asymptotic and the novelty status; it has not yet been completed.

References

[1] P. Grohs, *Continuous Shearlet Frames and Resolution of the Wavefront Set*, arXiv:0909.3712; Monatsh. Math. 164 (2011), 393–426.

[2] P. Grohs, *Continuous Shearlet Tight Frames*, arXiv:1001.1516; J. Fourier Anal. Appl. 17 (2011), 506–518.